

# Luis Alfredo Cuamatzi Flores

AI Engineer / Software Developer

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## SUMMARY

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AI Engineer and Software Developer with experience in mobile applications, backend services, and AI model integration. I have worked on real-world projects combining Android, APIs, automation, data processing, CRM, computer vision, and product-oriented interfaces to connect users, data, and business processes.

## EXPERIENCE

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### Software & AI Developer

Jun 2025 – Present

*Net2Alliance*

*Mexico City, Mexico*

- Full-stack development and AI integration for telecom and CRM platforms, working with FastAPI, Python, PHP, JavaScript, WebSockets, and AWS SageMaker.
- Built 2 modules and 2 endpoints for sentiment analysis and automated PDF summary generation from call transcripts using AWS SageMaker.
- Developed a WebSocket-based WhatsApp messaging client with real-time notifications, multimedia support, and agent transfer workflows.
- Implemented 4 CRM modules with 4 endpoints, CRUD operations, access control for 3 user roles, audit logging, and administrative reporting.
- Contributed to UI/UX redesigns with Figma and task tracking using Jira, GitHub, and Kanban boards.

### AI & Backend Engineer – Graduation Project / R&D

Jan 2025 – Dec 2025

*L4L – 3D Garment Reconstruction for Virtual Try-On*

*Tlaxcala, Mexico*

- Designed a client-server architecture for Virtual Try-On, integrating an Android app built with Jetpack Compose, a Linux inference server, and a Windows server for physical simulation and FBX-GLB conversion.
- Developed a FastAPI orchestrator to coordinate image capture, avatar generation and confirmation, garment reconstruction, physical simulation, and delivery of 3D models in GLB format.
- Integrated a single-image 2D body pose and shape estimation model (SMPLer-X), adapting its output for compatibility with the simulation pipeline.
- Adapted a two-level Transformer model for sewing pattern reconstruction from a 2D image (Sewformer), evaluating pruning, PTQ, and QAT variants with geometric and connectivity metrics.
- Compressed the final model with a 5.43x compression ratio and a 38.75% latency reduction, reaching 32.26 MB and 165.63 ms CPU inference time.

### Android Developer – Social Service

Oct 2024 – Aug 2025

*Instituto Politécnico Nacional – Institutional Tutoring Program*

*Tlaxcala, Mexico*

- Implemented QR-based attendance scanning with real-time validation, duplicate detection, and active session control.
- Built a 3-role user system (Admin, Peer Tutor, and Advisor) with separate panels based on permissions.
- Built a weekly schedule generator with overlap detection for classrooms and tutors.
- Implemented offline-first synchronization with Firestore and CSV export for institutional reporting.

## EDUCATION

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### B.Eng. in Artificial Intelligence Engineering

Aug 2021 – Jan 2026

*Instituto Politécnico Nacional (IPN) – UPIIT*

*Tlaxcala, Mexico*

### Leadership Formation Program

Aug 2024

*Queen Mary University of London*

*London, UK*

PROJECTS

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| <b>Lexfy</b><br><i>Android + AI Project</i> <ul style="list-style-type: none"><li>Android app with OCR and AI image generation through a chat interface.</li><li>Android client communicating via HTTP with a local Python/Flask backend running EasyOCR, GOT-OCR2_0, and Together AI.</li><li><b>Tech:</b> Kotlin, Jetpack Compose, Python, Flask, Firebase, EasyOCR, GOT-OCR2_0, Transformers.</li></ul>                                                                                                             | Dec 2024<br><a href="#">GitHub</a> |
| <b>SmartEMG Vision</b><br><i>Android prototype for assistive interaction</i> <ul style="list-style-type: none"><li>Mobile prototype combining EMG signal classification and object detection to suggest context-aware actions.</li><li>Android client with Jetpack Compose and CameraX sending data via HTTP to local Python servers running YOLOv8 and TensorFlow/Keras.</li><li><b>Tech:</b> Kotlin, Jetpack Compose, CameraX, Python, Flask, TensorFlow, Keras, YOLOv8.</li></ul>                                   | Nov 2024<br><a href="#">GitHub</a> |
| <b>SmartCrops</b><br><i>Mobile prototype for greenhouse monitoring</i> <ul style="list-style-type: none"><li>Prototype for crop monitoring, greenhouse variable simulation, and AI-based chatbot assistance (TomaBot, powered by Gemini).</li><li>Single-activity Jetpack Compose app with CameraX, OkHttp, and Coil.</li><li><b>Tech:</b> Kotlin, Jetpack Compose, Gemini API, CameraX, OkHttp, Coil.</li></ul>                                                                                                       | Aug 2024<br><a href="#">GitHub</a> |
| <b>CommuniSync</b><br><i>Android community platform</i> <ul style="list-style-type: none"><li>Built a native Android app for neighborhood engagement, including civic issue reports, polls, forums, a local directory, notifications, user profiles, and guest access.</li><li>Integrated Firebase Authentication and Firestore for email, Google, and guest sign-in, data storage, and real-time synchronization.</li><li><b>Tech:</b> Kotlin, Jetpack Compose, Material 3, Firebase Auth, Firestore, Coil.</li></ul> | May 2024<br><a href="#">GitHub</a> |
| <b>EduStream API</b><br><i>Web-based educational content management platform</i> <ul style="list-style-type: none"><li>Web platform for uploading, categorizing, and streaming educational videos by subject.</li><li>MVC backend with multimedia file management and synchronization between physical storage and the database.</li><li><b>Tech:</b> Node.js, Express, MySQL, Handlebars, Bootstrap, Multer.</li></ul>                                                                                                | Jun 2023<br><a href="#">GitHub</a> |

TECHNICAL SKILLS

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| <b>Languages:</b> Kotlin, Python, JavaScript, TypeScript, PHP, SQL                                                             |
| <b>Mobile:</b> Native Android, Jetpack Compose, Android Views/XML, Material 3, CameraX, Room, WorkManager, OkHttp, Coil, ZXing |
| <b>Backend:</b> FastAPI, Flask, Node.js, Express, WebSockets, REST APIs                                                        |
| <b>AI / ML:</b> TensorFlow, Keras, PyTorch, Transformers, EasyOCR, YOLOv8, SMPLer-X, Gemini API, Together AI                   |
| <b>Databases / Cloud:</b> AWS SageMaker, Firestore, Firebase Realtime Database, Firebase Storage, MySQL, SQLite                |
| <b>Architecture &amp; Methodologies:</b> Agile / Scrum, Kanban, MVC, MVVM, offline-first architecture, REST API design         |
| <b>Tools:</b> Git, GitHub, Jira, Figma, Postman, Android Studio, VS Code, Linux/CLI, SSH                                       |

LANGUAGES

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| <b>Spanish:</b> Native                                |
| <b>English:</b> B2 – Professional working proficiency |